

Chapter 3, Data Analysis, miscellaneous charts

This chapter graphs data from the 2016 datasets.

Import python libraries

```
In [16]: import pandas as pd
import numpy as np
import seaborn as sns
import matplotlib as mpl
from matplotlib import pyplot as plt
from collections import defaultdict

%matplotlib inline
```

Import the data that was created in the 'Data Transformation' workbook

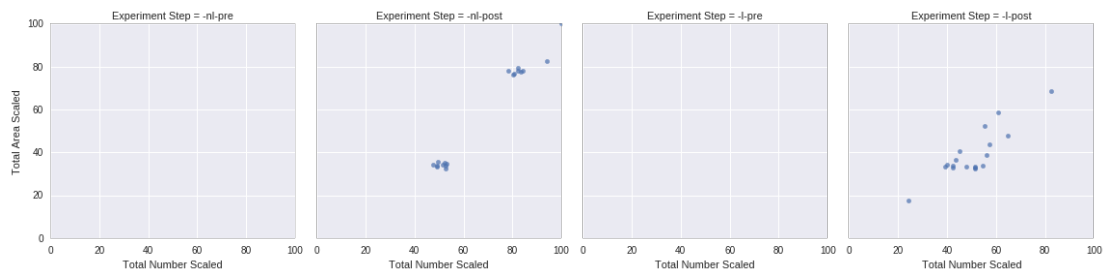
```
In [17]: totals65 = None
totals65 = pd.read_csv('T65_Totals.csv')

totals125 = None
totals125 = pd.read_csv('T125_Totals.csv')
```

Receptors 1 by 1, area by number, all steps

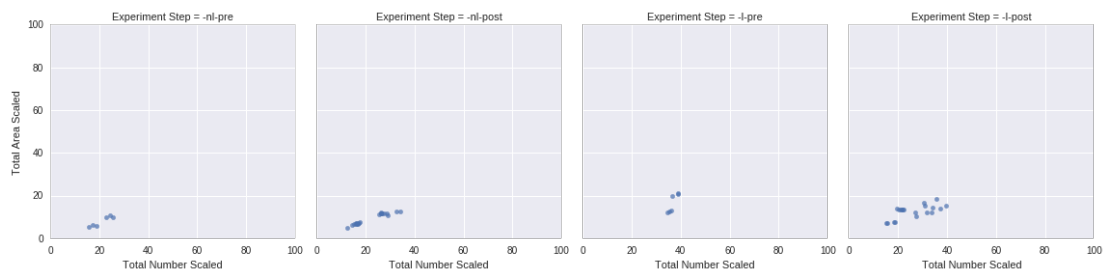
```
In [18]: tmpdf = totals65.loc[totals65['Receptor'] == 'A3']

g = sns.FacetGrid(tmpdf, col='Experiment Step', col_wrap=4,
                  size=4, ylim=(0,100), xlim=(0,100),
                  col_order=['-nl-pre', '-nl-post', '-l-pre', '-l-post'])
g.map(plt.scatter, 'Total Number Scaled', 'Total Area Scaled', alpha=.7)
g.savefig('total65_a3_by_steps.png')
```



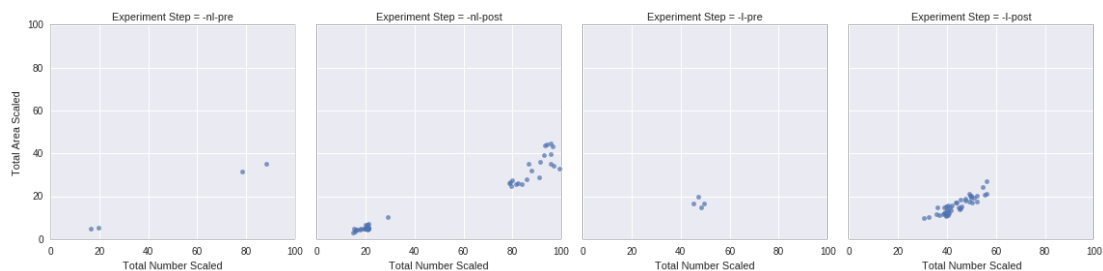
```
In [19]: tmpdf = totals65.loc[totals65['Receptor'] == 'A5']

g = sns.FacetGrid(tmpdf, col='Experiment Step', col_wrap=4,
                  size=4, ylim=(0,100), xlim=(0,100),
                  col_order=['-nl-pre', '-nl-post', '-l-pre', '-l-post'])
g.map(plt.scatter, 'Total Number Scaled', 'Total Area Scaled', alpha=.7)
g.savefig('total65_a5_by_steps.png')
```



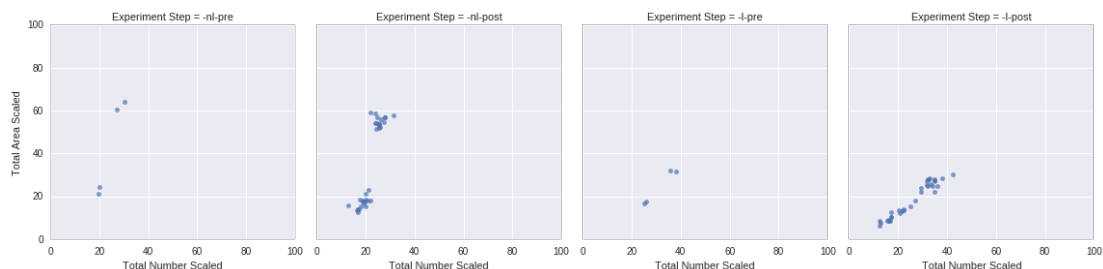
```
In [20]: tmpdf = totals65.loc[totals65['Receptor'] == 'B1']

g = sns.FacetGrid(tmpdf, col='Experiment Step', col_wrap=4,
                  size=4, ylim=(0,100), xlim=(0,100),
                  col_order=['-nl-pre', '-nl-post', '-l-pre', '-l-post'])
g.map(plt.scatter, 'Total Number Scaled', 'Total Area Scaled', alpha=.7)
g.savefig('total65_b1_by_steps.png')
```



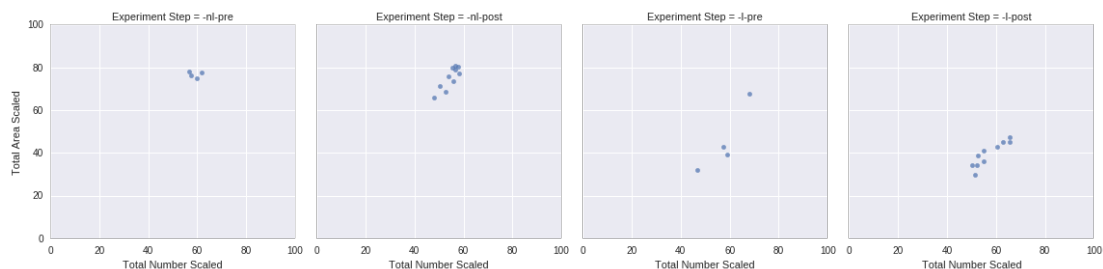
```
In [21]: tmpdf = totals65.loc[totals65['Receptor'] == 'M1']

g = sns.FacetGrid(tmpdf, col='Experiment Step', col_wrap=4,
                  size=4, ylim=(0,100), xlim=(0,100),
                  col_order=['-nl-pre', '-nl-post', '-l-pre', '-l-post'])
g.map(plt.scatter, 'Total Number Scaled', 'Total Area Scaled', alpha=.7)
g.savefig('total65_m1_by_steps.png')
```



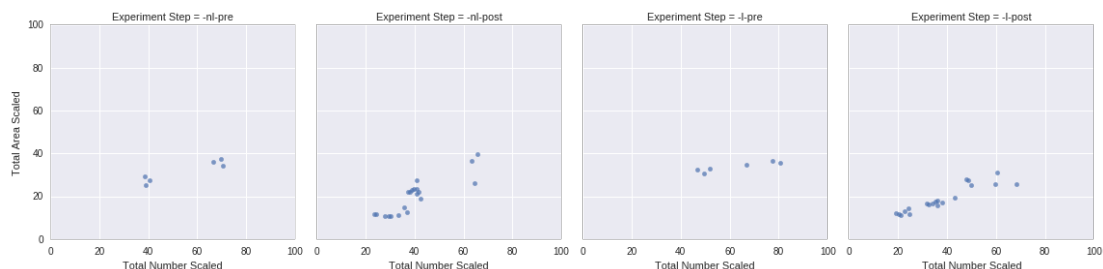
```
In [22]: tmpdf = totals65.loc[totals65['Receptor'] == 'M7']

g = sns.FacetGrid(tmpdf, col='Experiment Step', col_wrap=4,
                  size=4, ylim=(0,100), xlim=(0,100),
                  col_order=['-nl-pre', '-nl-post', '-l-pre', '-l-post'])
g.map(plt.scatter, 'Total Number Scaled', 'Total Area Scaled', alpha=.7)
g.savefig('total65_m7_by_steps.png')
```



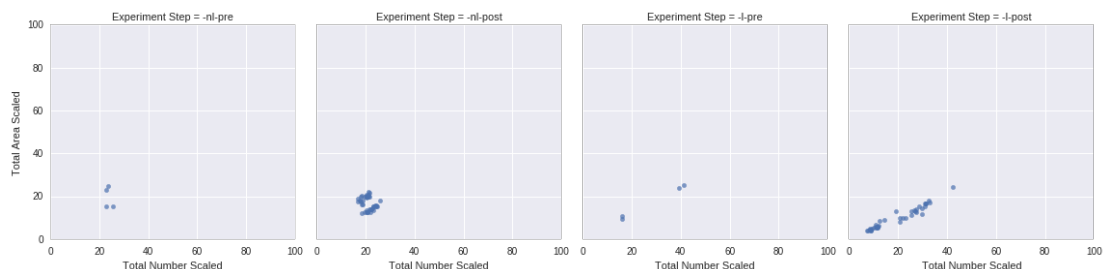
```
In [23]: tmpdf = totals65.loc[totals65['Receptor'] == 'M22']

g = sns.FacetGrid(tmpdf, col='Experiment Step', col_wrap=4,
                  size=4, ylim=(0,100), xlim=(0,100),
                  col_order=['-nl-pre', '-nl-post', '-l-pre', '-l-post'])
g.map(plt.scatter, 'Total Number Scaled', 'Total Area Scaled', alpha=.7)
g.savefig('total65_m22_by_steps.png')
```



```
In [24]: tmpdf = totals65.loc[totals65['Receptor'] == 'M26']

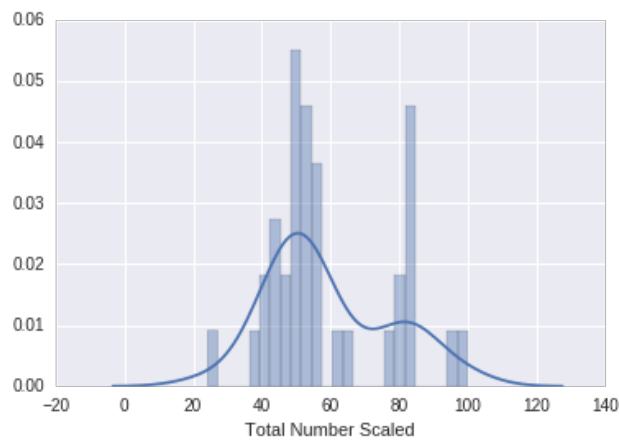
g = sns.FacetGrid(tmpdf, col='Experiment Step', col_wrap=4,
                  size=4, ylim=(0,100), xlim=(0,100),
                  col_order=['-nl-pre', '-nl-post', '-l-pre', '-l-post'])
g.map(plt.scatter, 'Total Number Scaled', 'Total Area Scaled', alpha=.7)
g.savefig('total65_m26_by_steps.png')
```



Receptors 1 by 1, number, kernel density smoothing on histograms

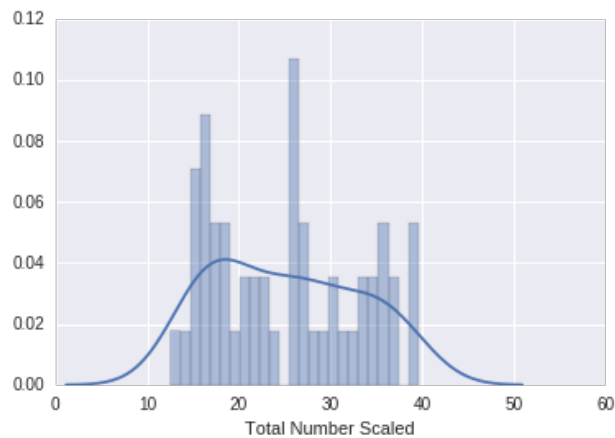
```
In [25]: tmpdf = totals65.loc[totals65['Receptor'] == 'A3']

sns_plot = sns.distplot(tmpdf['Total Number Scaled'], kde=True, bins=25)
fig = sns_plot.get_figure()
fig.savefig('total65_a3_by-number_histogram-1.png')
```



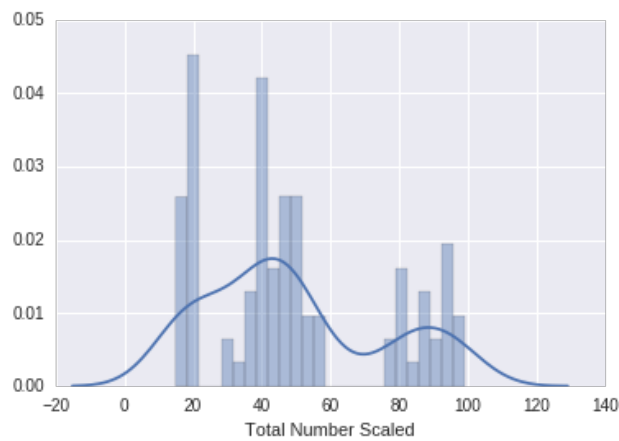
```
In [26]: tmpdf = totals65.loc[totals65['Receptor'] == 'A5']

sns_plot = sns.distplot(tmpdf['Total Number Scaled'], kde=True, bins=25)
fig = sns_plot.get_figure()
fig.savefig('total65_a5_by-number_histogram-1.png')
```



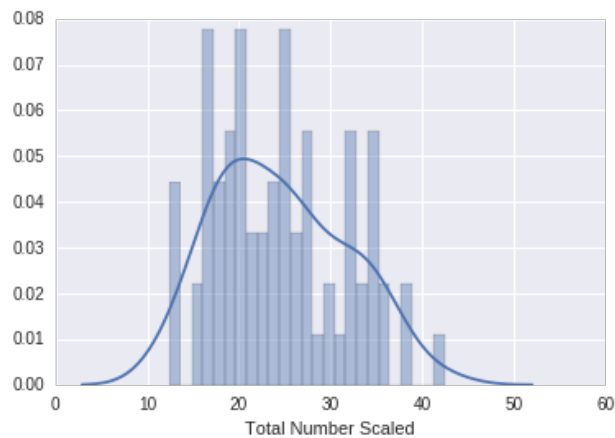
```
In [27]: tmpdf = totals65.loc[totals65['Receptor'] == 'B1']

sns_plot = sns.distplot(tmpdf['Total Number Scaled'], kde=True, bins=25)
fig = sns_plot.get_figure()
fig.savefig('total65_b1_by-number_histogram-1.png')
```



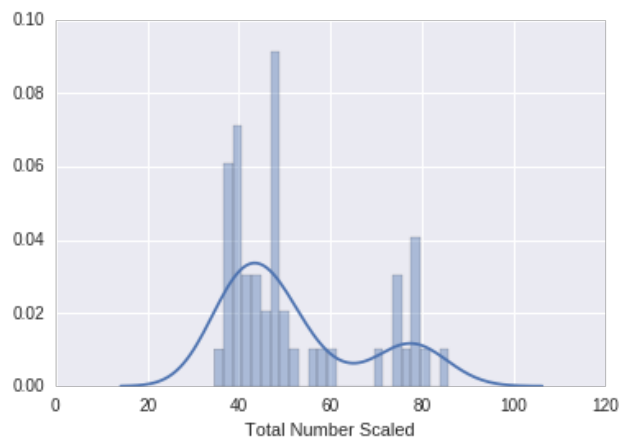
```
In [28]: tmpdf = totals65.loc[totals65['Receptor'] == 'M1']

sns_plot = sns.distplot(tmpdf['Total Number Scaled'], kde=True, bins=25)
fig = sns_plot.get_figure()
fig.savefig('total65_m1_by-number_histogram-1.png')
```



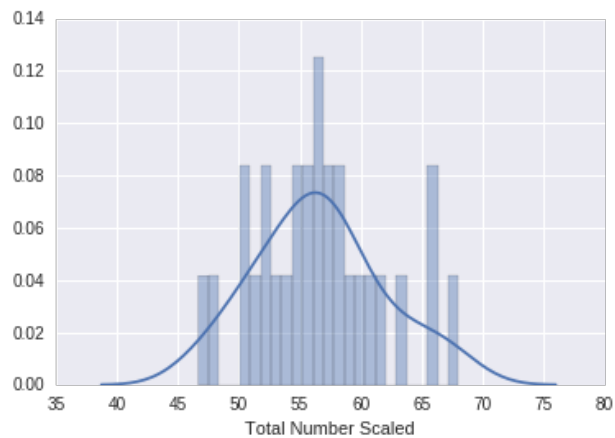
```
In [29]: tmpdf = totals65.loc[totals65['Receptor'] == 'M5']

sns_plot = sns.distplot(tmpdf['Total Number Scaled'], kde=True, bins=25)
fig = sns_plot.get_figure()
fig.savefig('total65_m5_by-number_histogram-1.png')
```



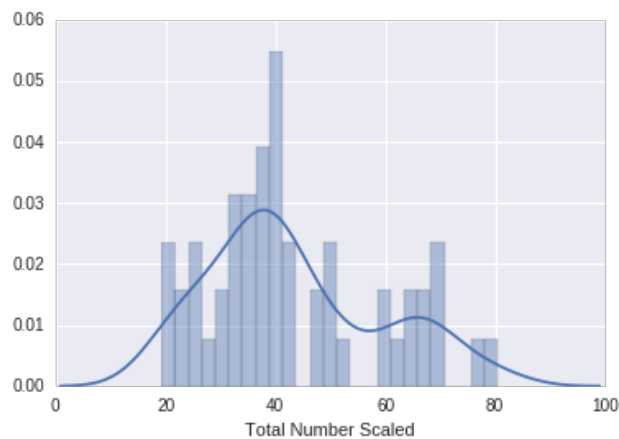
```
In [30]: tmpdf = totals65.loc[totals65['Receptor'] == 'M7']

sns_plot = sns.distplot(tmpdf['Total Number Scaled'], kde=True, bins=25)
fig = sns_plot.get_figure()
fig.savefig('total65_m7_by-number_histogram-1.png')
```



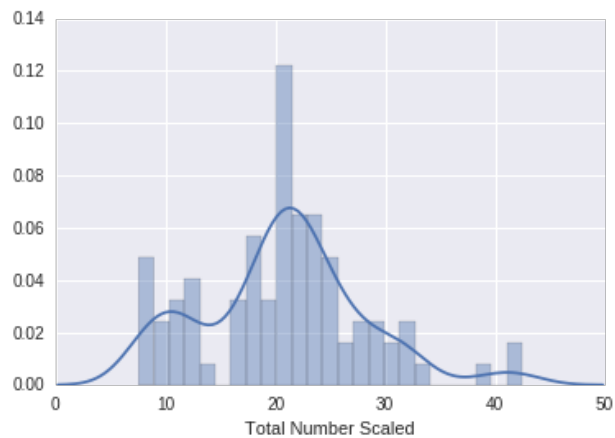
```
In [31]: tmpdf = totals65.loc[totals65['Receptor'] == 'M22']

sns_plot = sns.distplot(tmpdf['Total Number Scaled'], kde=True, bins=25)
fig = sns_plot.get_figure()
fig.savefig('total65_m22_by-number_histogram-1.png')
```



```
In [32]: tmpdf = totals65.loc[totals65['Receptor'] == 'M26']

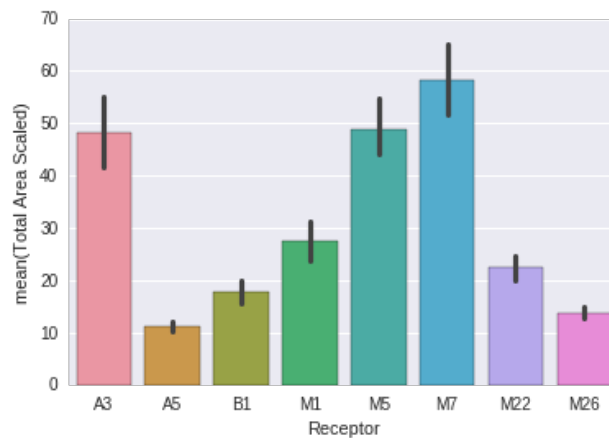
sns_plot = sns.distplot(tmpdf['Total Number Scaled'], kde=True, bins=25)
fig = sns_plot.get_figure()
fig.savefig('total65_m26_by-number_histogram-1.png')
```



All receptors, area, mean size histogram

```
In [33]: tmpdf = totals65.loc[(totals65['Receptor'].isin(['M1', 'M5', 'M7', 'M22', 'M26'])) &
                                (totals65['Experiment Step'] == '-nl-post')]

g = sns.barplot(x=totals65['Receptor'],
                y=totals65['Total Area Scaled'],
                order=['A3', 'A5', 'B1', 'M1', 'M5', 'M7', 'M22', 'M26'])
plt.savefig('total65_hist_total_area_nl-post.png')
```



All receptors, area, mean number histogram

```
In [34]: tmpdf = totals65.loc[(totals65['Receptor'].isin(['M1', 'M5', 'M7', 'M22', 'M26'])) &
                                (totals65['Experiment Step'] == '-nl-post')]

g = sns.barplot(x=totals65['Receptor'],
                y=totals65['Total Number Scaled'],
                order=['A3', 'A5', 'B1', 'M1', 'M5', 'M7', 'M22', 'M26'])
plt.savefig('total65_hist_total_number_nl-post.png')
```

